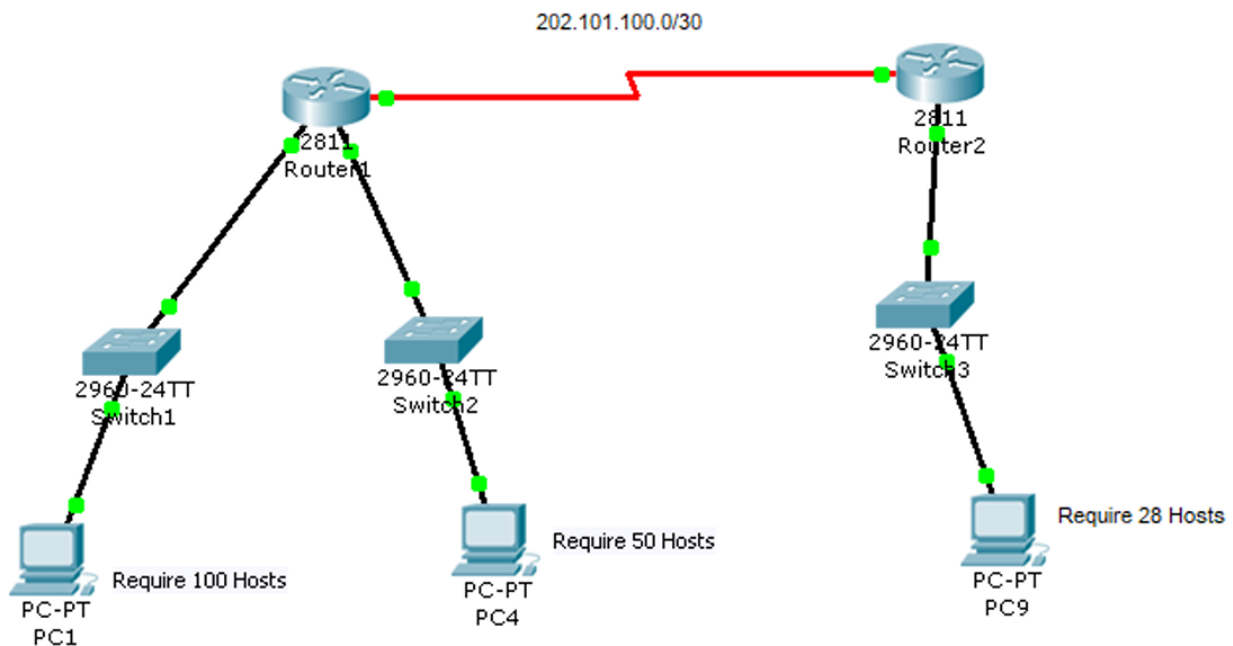




Task1, IP Belongs to which Class:

Question: Use 202.101.100.0 & select a class in which the requirement satisfy. What class should be used to accomplish this topology?

Answer: This IP belongs to Class "C".

Task 2, Requirement

Router 1&2 (Se0/0/0): Require 2 useable hosts

Router 1(Fa0/0) Require 100 Useable Hosts

Router 1(Fa0/1) Require 50 usable Hosts

Router 2(Fa0/0) Require 28 usable Hosts

Task 3, Fill the table to satisfy the requirements.

Device Name	Interface	IP address Range	Subnet Mask	Default Gateway
Router 1	Se 0/0/0	202.101.100.225 - 202.101.100.226	255.255.255.252	202.101.100.225
Router 1	Fa 0/0	202.101.100.1 - 202.101.100.126	255.255.255.128 (/25)	202.101.100.1
Router 1	Fa 0/1	202.101.100.129 - 202.101.100.190	255.255.255.192 (/26)	202.101.100.129
Router 2	Se 0/0/0	202.101.100.229 - 202.101.100.230	255.255.255.252	202.101.100.229
Router 2	Fa 0/0	202.101.100.193 - 202.101.100.222	255.255.255.224 (/27)	202.101.100.193

$$2^7 = 128$$

$$2^7 - 2 = 128 - 2 = 126$$

$$32 - 7 = 25$$

$$\text{Network address} = 202.101.100.0 + 1 / 25$$

$$\text{First address} = 202.101.100.1 \rightarrow 125$$

$$\text{Last address} = \text{Network} + \text{valid host} = 202.101.100.126$$

$$\text{Broadcast Address} = 202.101.100.127 \quad 125$$

Subnet mask

8

8

8

8

$$32 - 25 = 7$$

$$8 = 32$$

11111111

11111111

11111111

10000000

255

255

255

128

Broadcast + 1 = current IP address

80 host

Network = 202.101.100.128/26

FA = 202.101.100.129/26

LA = 202.101.100.190/26

BA = 202.101.100.191/26

 $2^6 = 64$ Address $2^6 - 2 = 62$ valid host $32 - 6 = 26$ subnet

Subnet Mask:

8	8	8	8	$32 - 26 = 6$
11111111	11111111	11111111	11111111	$= 32$
255	255	255	255	192

28 Host

Network Address: 202.101.100.192/27

FA: 202.101.100.193/27

LA: 202.101.100.222/27

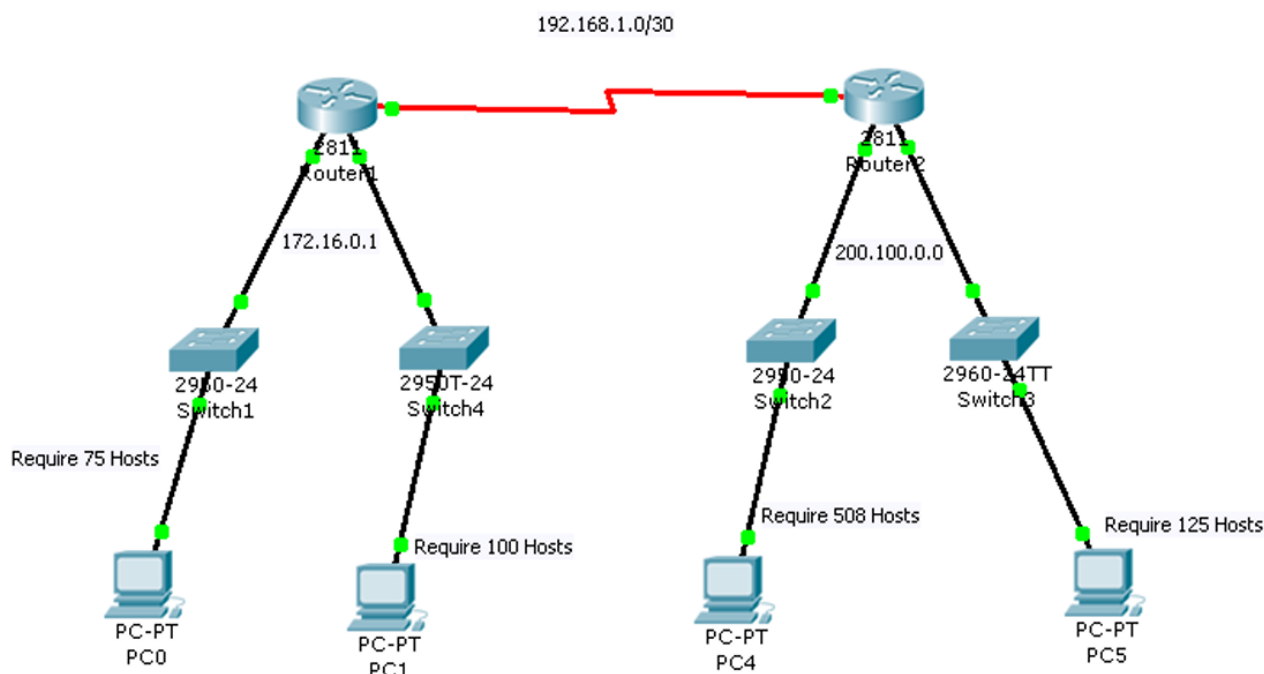
BA: 202.101.100.223/27

 $2^5 = 32$ Address $2^5 - 2 = 30$ valid host $32 - 5 = 27$ subnet mask

Subnet mask

8	8	8	8	$32 - 5 = 27$
11111111	11111111	11111111	11111111	
255	255	255	224	

255.255.255.224



Requirement

Question: Use the above IP Addresses & select a class in which the requirement satisfy What class should be used for Router 1 Interfaces? Answer: Class B

Question: What class should be used for Router 1 & 2 Serial Interfaces? Answer: Class C

Question: What class should be used for Router 2 interfaces? Answer: Class C

Router 1&2 (Se0/0/0): Require 2 useable hosts

Router 1(Fa0/0) Require 75 Useable Hosts

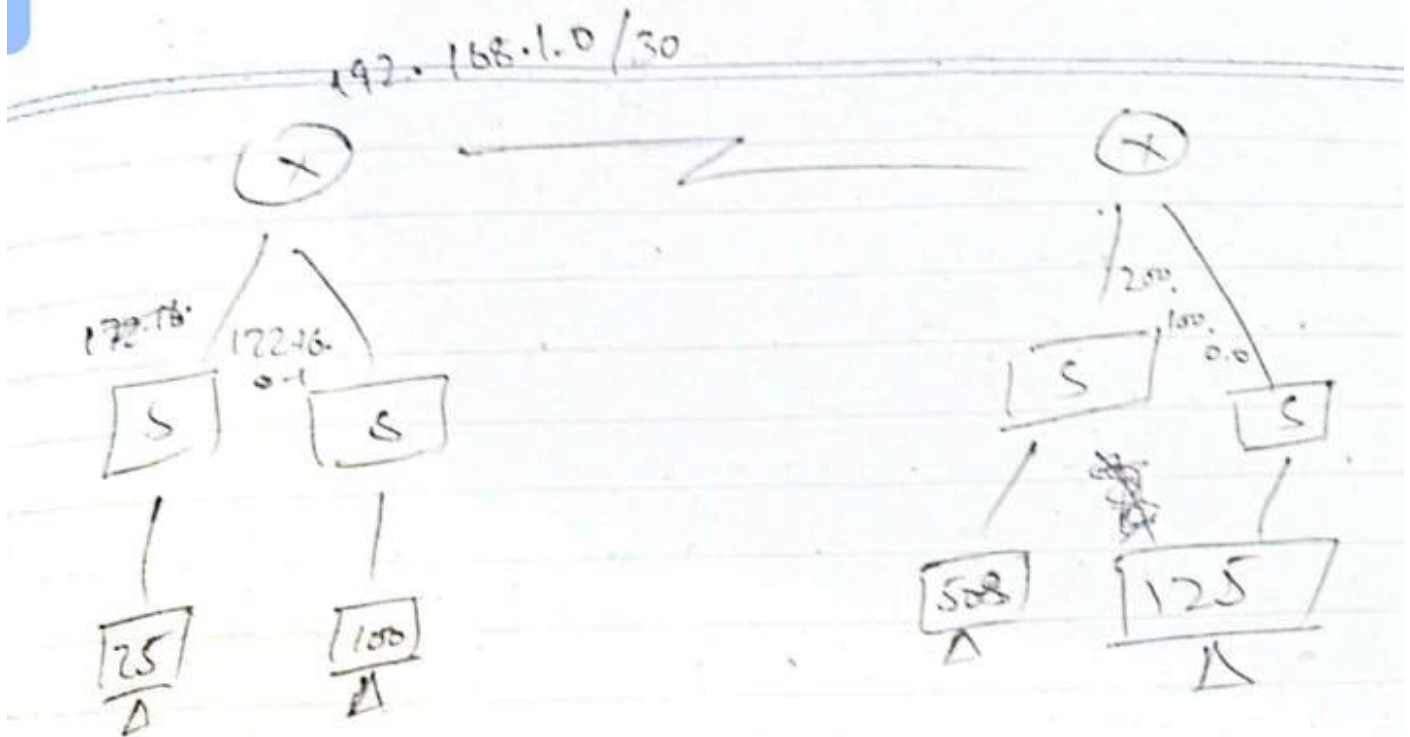
Router 1(Fa0/1) Require 100 usable Hosts

Router 2(Fa0/0) Require 508 usable Hosts

Router 2(Fa0/1) Require 125 usable hosts

Fill the table to satisfy the requirements

Device Name	Interface	IP Addresses (Range)	Subnet	Default Gateway
Router 1	Se0/0/0	192.168.1.1 - 192.168.1.2	255.255.255.252 (/30)	192.168.1.1
Router 1	Fa0/0	172.16.0.129 - 172.16.0.254	255.255.255.128 (/25)	172.16.0.129
Router 1	Fa0/1	172.16.0.1 - 172.16.0.126	255.255.255.128 (/25)	172.16.0.1
Router 2	Se0/0/0	192.168.1.5 - 192.168.1.6	255.255.255.252 (/30)	192.168.1.5
Router 2	Fa0/0	200.100.0.1 - 200.100.1.254	255.255.254.0 (/23)	200.100.0.1
Router 2	Fa0/1	200.100.2.1 – 200.100.2.126	255.255.255.128 (/25)	200.100.2.1



508 Hosts

$$2^9 = 512$$

$$2^9 - 2 = 510$$

$$32 - 9 = 23$$

$$N = 200.100.0.0 / 23 \quad 510/2 = \underline{255} \quad N$$

$$FA = 200.100.0.1 / 23$$

~~$$LA = 200.100.255.255$$~~

$$LA = 200.100.1.254 / 23$$

$$BC = 200.100.1.255 / 23$$

subnet

$$32 - 23$$

$$\underline{255.255.254.0}$$

125 Host

$$2^7 = 128$$

$$2^7 - 2 = 126$$

$$32 - 7 = 25$$

Network = 200.100.1.

$$N = 200.100.2.0 / 25$$

$$A = 200.100.2.1 / 25$$

$$LA = 200.100.2.126 / 25$$

$$BC = 200.100.2.127 / 25$$

Subnet

$$32 - 25 = 7$$

1111111 1111111

255.255.255.128

100 Hosts

$$2^7 = 128 \text{ (IP address)}$$

$$2^7 - 2 = 126 \text{ (valid host)}$$

$$32 - 7 = 25 \text{ (subnet)}$$

172.16.0.1

Network address: 172.16.0.0 / 25

$$PA = 172.16.0.1 / 25$$

$$LA = 172.16.0.126 / 25$$

$$BC = 172.16.0.127 / 25$$

$$1 + 126 = 127$$

Subnet Mask $32 - 25 = 7$

1111111.1111111.1111111.10000000

10 Hosts

255

$$2^7 = 128$$

$$2^7 - 2 = 126$$

$$32 - 7 = 25$$

$$Netw 172.16.0.128 / 25$$

$$PA = 172.16.0.129 / 25$$

$$LA = 172.16.0.254$$

$$BC = 172.16.0.255$$

$$LA = 172.16.0.254 / 25$$

$$BA = 172.16.0.$$

$$BA = 172.16.0.255 / 25$$

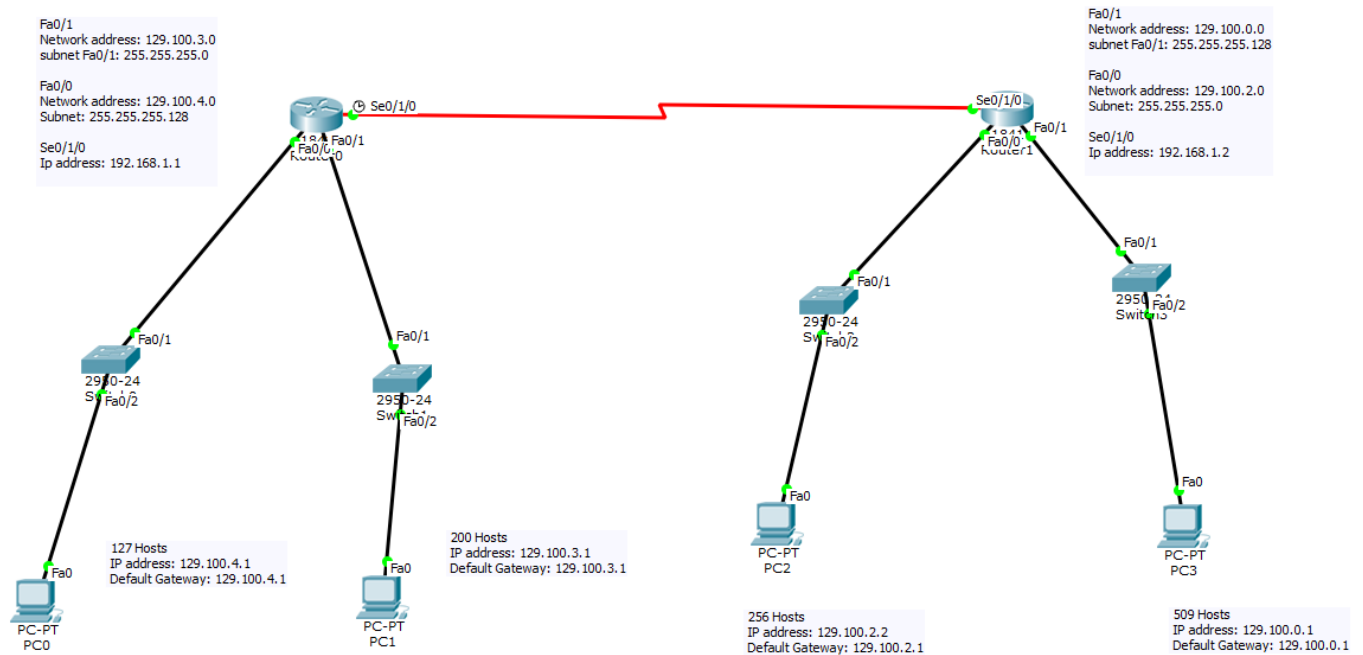
Subnet

$$32 - 25 = 7$$

1111111.1111111.1111111.10000000

255.255.255.128

② 4



which the requirement satisfy What class should be used for Router 1 Interfaces?

Class B

What class should be used for Router 1 & 2 Serial Interfaces?

Class C

What class should be used for Router 2 interfaces?

Class B

509 Hosts

$$2^9 = 512 \quad \text{Ip addresses}$$

$$2^9 - 2 = 510 \quad \text{Valid hosts}$$

$$32 - 9 = 23 \quad \text{Subnet 'n'}$$

$$510/2 \begin{matrix} \nearrow 255 \\ \searrow 255 \end{matrix}$$

Network address: 129.100.0.0 /23

First Address: 129.100.0.1 /23

Last Address: 129.100.1.254 /23

Broad Cast Address: 129.100.1.255 /23

Subnet: 11111111.11111111.11111111.10000000

255.255.255.128

256 Hosts

$$2^8 = 256$$

$$2^8 - 2 = 254$$

$$32 - 8 = 24$$

Network Address: ~~129.100.0.0~~ 129.100.2.0 /24

First Address: 129.100.2.1 /24

Last Address: 129.100.2.254 /24

Broad Cast Address: 129.100.2.255 /24

Subnet: 11111111.11111111.11111111.00000000

255.255.255.0

200 Host

$$2^8 = 256$$

$$2^8 - 2 = 254$$

$$32 - 8 = 24$$

Network Address: 129.100.3.0 / 24

First Address: 129.100.3.1 / 24

Last Address: 129.100.3.254 / 24

Broad Cast Address: 129.100.3.255 / 24

Subnet: 11111111.11111111.11111111.00000000

255.255.255.0

127 Host

$$2^7 = 128$$

$$2^7 - 2 = 126$$

$$32 - 7 = 25$$

Network Address: 129.100.4.0 / 25

First Address: 129.100.4.1 / 25

Last Address: 129.100.4.126 / 25

Broad Cast Address: 129.100.4.127 / 25

Subnet: 11111111.11111111.11111111.10000000

255.255.255.128

